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HUJI Applicative Research Report — Week 43, 2026

2 high-potential paper(s) · Weekly · Week 43, 2026 · Generated October 19, 2026

Both papers highlight innovative approaches in drug discovery and vaccine development from Dr. Netanel Tzarum's lab. One groundbreaking platform technology enables the safe and scalable production of highly toxic bacterial proteins, opening vast commercial avenues for next-generation antibody-drug conjugates and novel vaccines. Concurrently, new structural insights into Hepatitis C Virus envelope proteins offer a crucial foundation for designing a much-needed first-in-class preventative vaccine against the global health burden of HCV. These advancements present compelling investment opportunities in novel therapeutic and prophylactic strategies.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Netanel Tzarum** — Drug Discovery, Vaccines, Synthetic Biology, Immunology
- **Netanel Tzarum** — Vaccines, Immunology, Drug Discovery

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Modular Platform for Recombinant Production of Highly Toxic Bacterial Proteins. 37/50

"Breakthrough platform unlocks safe, high-yield production of highly toxic proteins for next-gen ADCs and novel vaccines."

Main researcher: Netanel Tzarum

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, Faculty of Mathematics & Science, The Hebrew University of Jerusalem, Jerusalem, Israel.; The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Microbial biotechnology · 2026 Oct 17

ABOUT THIS RESEARCH

This platform enables the safe, modular recombinant production of highly lethal bacterial proteins that are typically impossible to express using standard industrial methods. By decoupling the protein synthesis process from cellular viability, it allows for the high-yield manufacturing of potent toxins without endangering the production host.

COMMERCIAL ANGLE

The technology unlocks a massive commercial pipeline for next-generation antibody-drug conjugates (ADCs) and novel vaccines that leverage bacterial toxins previously deemed too dangerous or difficult to produce.

COMMERCIALISATION METRICS

Novelty 7/10

The development of a modular platform for toxic protein expression addresses a critical bottleneck in biotechnology and biodefense, offering high utility for industrial and therapeutic applications. While the approach represents a significant technical refinement for difficult-to-express proteins, it is an evolutionary improvement on existing protein engineering workflows rather than a fundamental paradigm shift in synthetic biology.

Commercial Potential 9/10

This platform addresses a high-value technical bottleneck in the production of potent biotherapeutics and toxins, offering a clear, scalable B2B model for contract manufacturing or licensing to pharmaceutical companies. By abstracting the production of toxic proteins into a modular system, it de-risks a traditionally difficult manufacturing process, providing strong defensibility through proprietary technology.

Market Size 7/10

The platform addresses a high-value niche in the biopharma and defense sectors by enabling the production of previously unmanufacturable toxic proteins, which are essential for targeted cancer therapies (ADC payloads) and vaccine development. While the addressable market for these specific high-toxicity biologics is smaller than broad-spectrum protein therapeutics, the technical barriers to entry confer significant strategic value and pricing power.

Tech Readiness 7/10

The development of a modular production platform for toxic proteins addresses a significant industrial bottleneck in biomanufacturing, demonstrating high technical utility and clear scalability potential. However, its TRL is moderated by the inherent safety, regulatory, and containment complexities associated with scaling the production of hazardous biological agents.

IP Strength 7/10

The platform approach for expressing difficult, toxic proteins offers strong utility and a clear path to composition-of-matter or process-based IP, though defensibility depends heavily on the novelty of the modular genetic control elements used to mitigate toxicity.

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2. Structural and biochemical insights into HCV envelope proteins for germline-targeting vaccines. 28/50

"Foundational insights pave the way for a first-in-class Hepatitis C vaccine, addressing a significant global health crisis."

Main researcher: Netanel Tzarum

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, Faculty of Mathematics & Science, The Hebrew University of Jerusalem, Jerusalem, Israel.; The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in PLoS pathogens · 2026 Oct 12

ABOUT THIS RESEARCH

This research explores the molecular architecture of Hepatitis C Virus (HCV) envelope proteins to identify specific targets for the immune system. By mapping these structures, the scientists aim to design vaccine candidates that can elicit broadly neutralizing antibodies across diverse viral strains.

COMMERCIAL ANGLE

The structural insights provide a platform for developing a first-in-class HCV vaccine, addressing a significant global health gap in preventing chronic liver disease.

COMMERCIALISATION METRICS

Novelty 6/10

While applying structural biology to the challenging HCV envelope glycoprotein is high-quality scientific work, the 'germline-targeting' strategy represents an incremental advancement of existing vaccine design paradigms rather than a novel conceptual breakthrough.

Commercial Potential 6/10

While the research targets a major global health challenge with clear licensing potential for pharmaceutical pipelines, it remains in the structural/biochemical phase, which is high-risk and years away from a de-risked clinical asset. The work is foundational and scientific rather than translational or near-market, limiting its immediate commercial clarity.

Market Size 7/10

Hepatitis C affects approximately 58 million people globally, representing a significant unmet medical need for a preventative vaccine that could command a large, multi-billion dollar addressable market. While the curative potential of current direct-acting antivirals limits the immediate urgency for a vaccine, the lack of a prophylactic option for high-risk populations keeps the market opportunity substantial.

Tech Readiness 3/10

This study focuses on foundational structural biology and biochemical characterization of viral proteins, which represents a discovery-stage effort far removed from clinical trial readiness or industrial manufacturing. While it provides essential data for vaccine design, it remains at the early-stage research level of the R&D pipeline.

IP Strength 6/10

While the structural characterization of viral proteins provides a foundational basis for rational vaccine design, the resulting insights often constitute scientific discovery rather than proprietary composition-of-matter inventions, necessitating significant downstream effort to secure specific vaccine candidates or delivery platforms. The inherent difficulty in achieving broad-spectrum efficacy against highly variable targets like HCV further complicates the establishment of a robust, enforceable IP moat.

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